

Thermal non-equilibrium (TNE)

Key process for the mass and energy transport in the solar corona

- ★ Set by a stratified and quasi-steady heating that will induce evaporation and condensation cycles e.g. Antiochos and Klimchuk, 1991
- ★ Global behaviour that connect many different scales (temporal, spatial, temperature and density conditions) and all the solar atmospheric layers

Widespread in the solar corona, manifestations includes:

- ➔ **Long-period EUV intensity pulsations** (periods of a few hours): reflecting the temperature and density evolution in the corona Auchère et al. 2014; Froment et al. 2015, 2017, 2020; Auchère et al. 2018; Pelouze et al. 2020; Sahin et al. 2023
- ➔ **Coronal rain**: cool and dense material, observed in chromospheric and transition-region lines, produced by a thermal instability e.g. Kawaguchi 1970, Leroy 1972, De Groof et al. 2004, Antolin & Rouppe van der Voort 2012

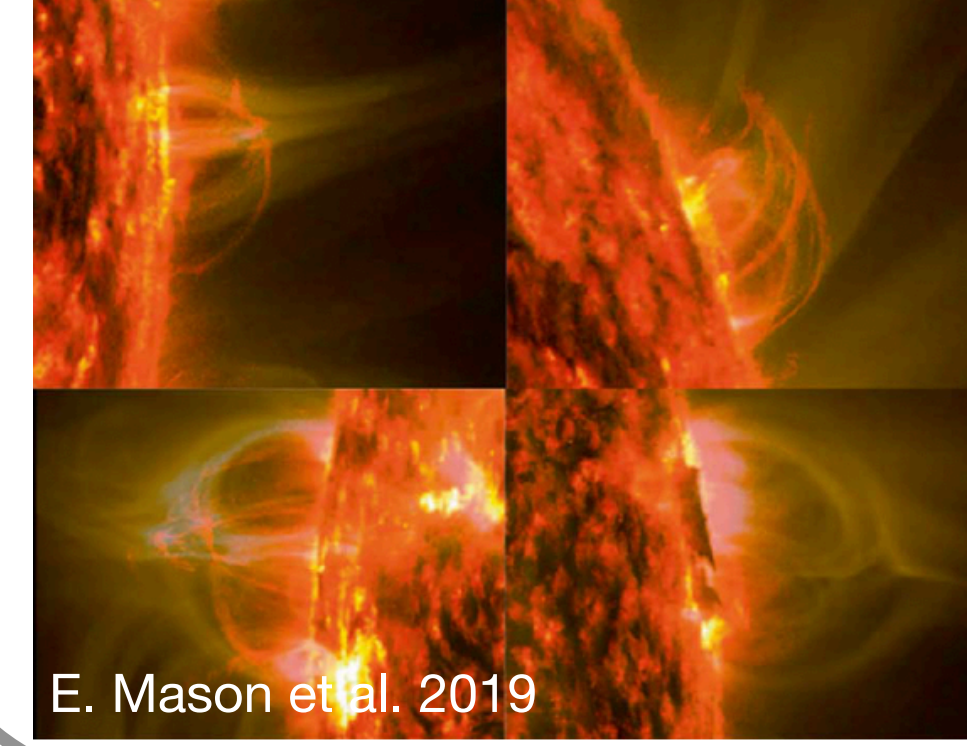
Is TNE restricted to closed fields?

Implications relatively well-known for closed-field structures, what about TNE at open-closed boundaries and in open structures?

Recent numerical works in open field structures

- transient coronal rain formed in open field Kohutova et al. 2020
- TNE cycles in open field Scott et al. 2024, Le Nestour et al. in prep

Relationship of these events with TNE to be investigated



Hybrid prominence-coronal-rain events

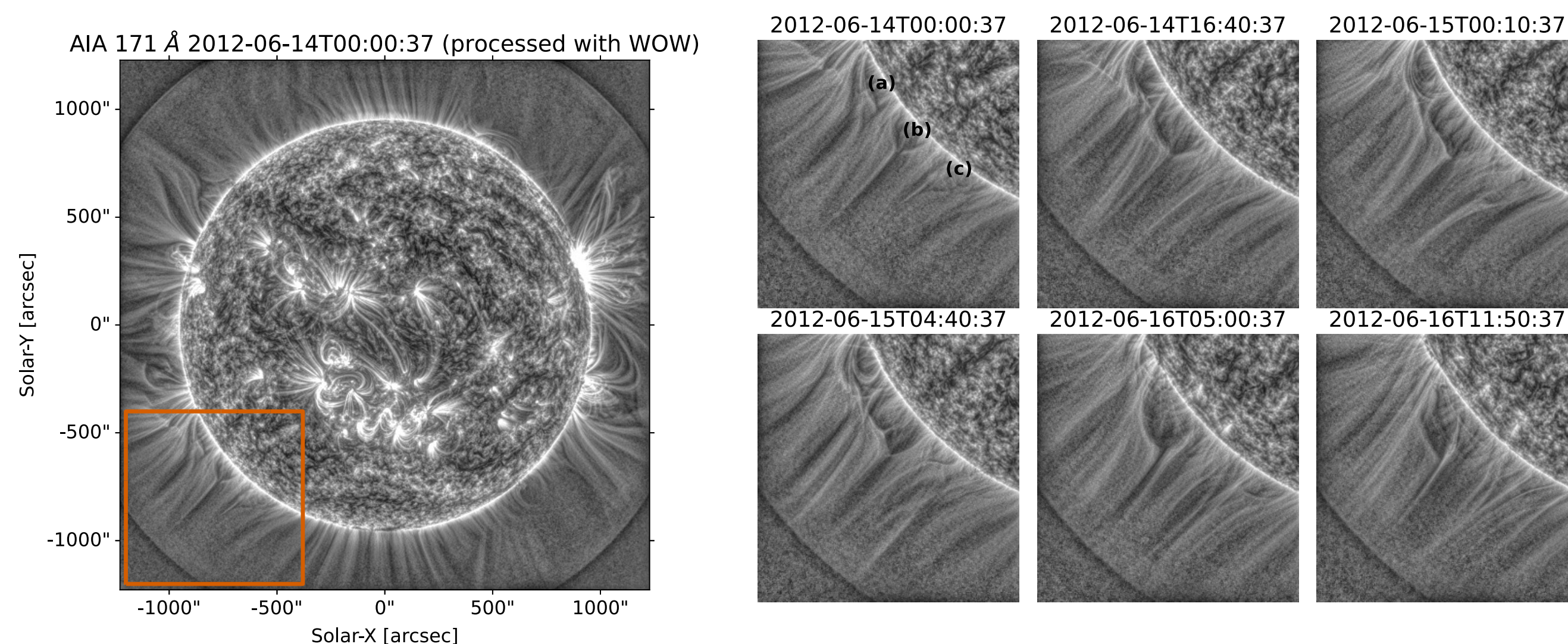
(Liu et al. 2014, Antolin & Froment 2022)

- ➔ Common, recurrent phenomenon in the quiet Sun Li et al. 2018, 2019, 2020, Mason et al. 2019
- ➔ Topological structures: Fan-spine topologies and pseudo-streamers

Mason et al. 2019 proposed interchange reconnection and TNE for the formation mechanism of these events

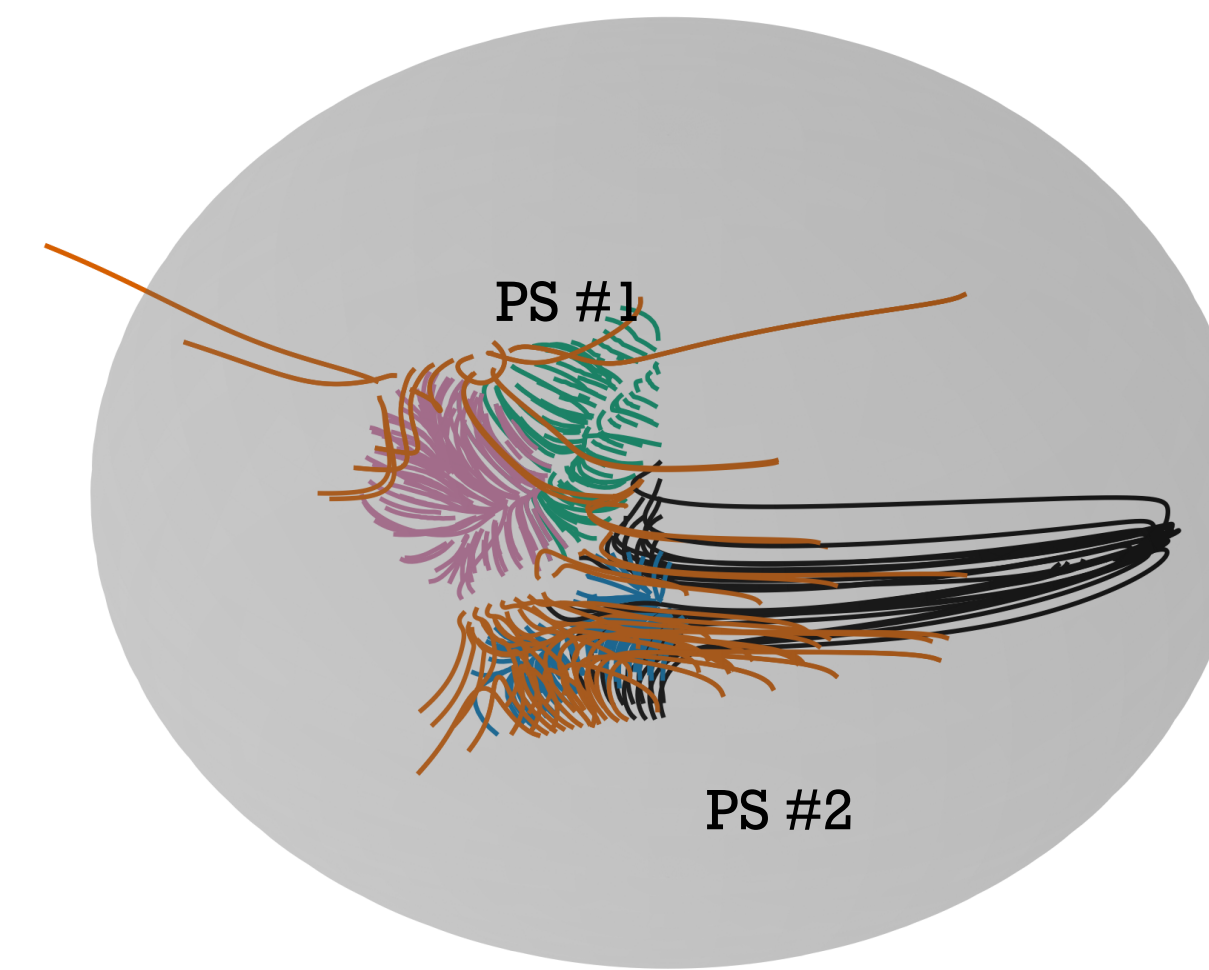
Case study in a pseudo-streamer (PS)

Global evolution in EUV (SDO/AIA)



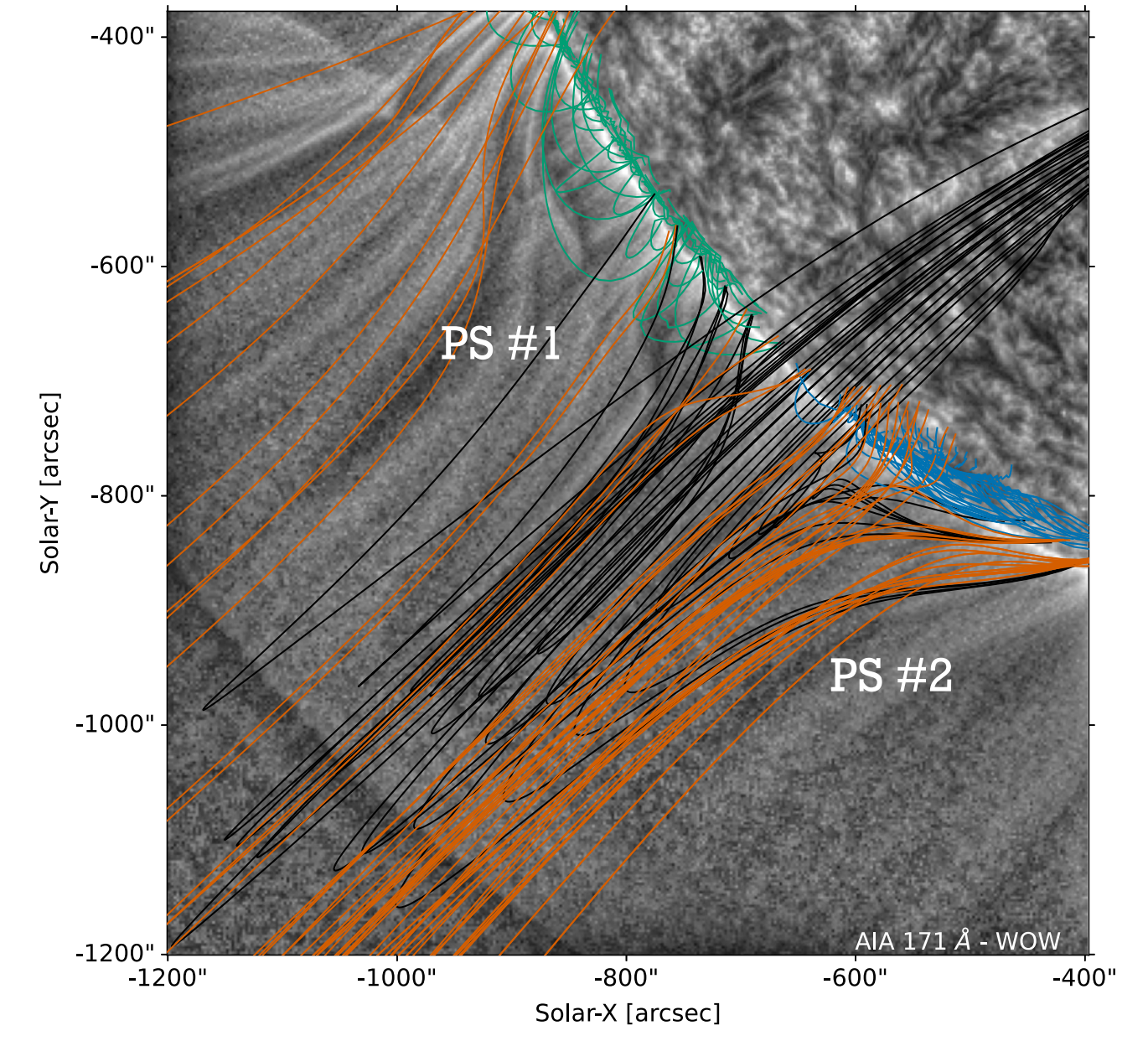
3 EUV lobes (?) => the northern and the middle lobes are in reality from the same PS-dome but with a change in its orientation.

Magnetic topology (PFSS extrapolations)



Pseudo-streamer:

- ➔ closed field lines confined by a dome separatrix surface
- ➔ an open vertical fan or separatrix curtain, located at the interface between two open-field regions of the same polarity. Wang et al. 2007; Titov et al. 2011, 2012; Scott et al. 2021

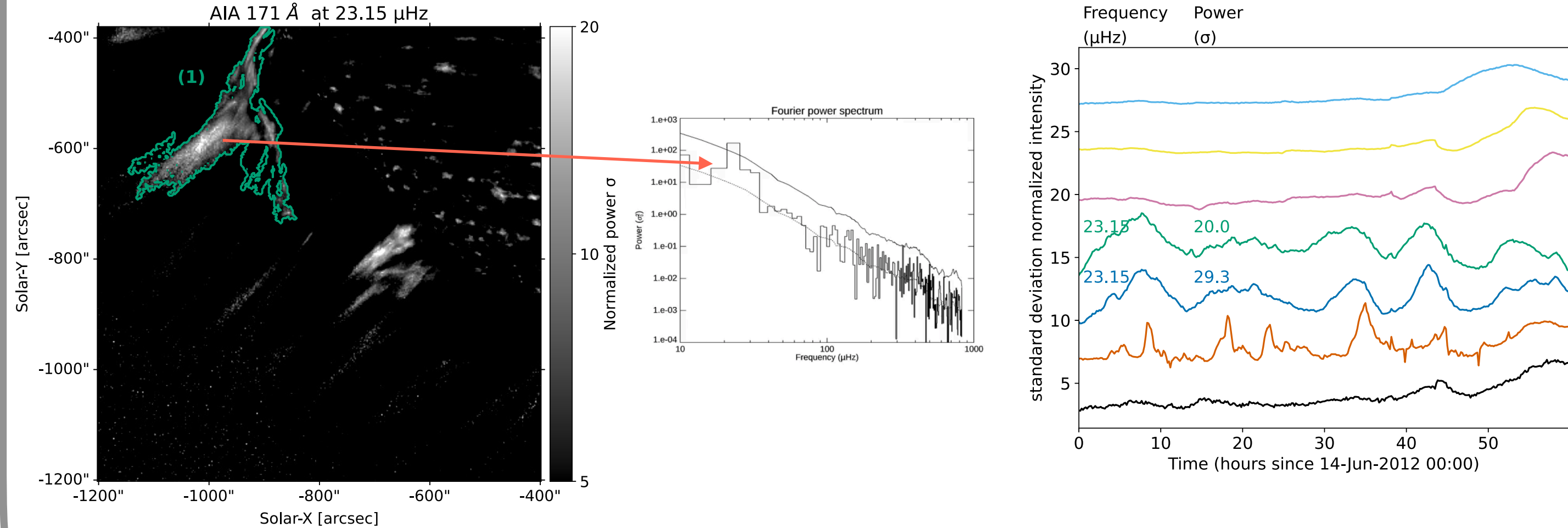


- ➔ PS #1 is the focus of our study: non-eruptive pseudo-streamers

Long-period EUV pulsations detections

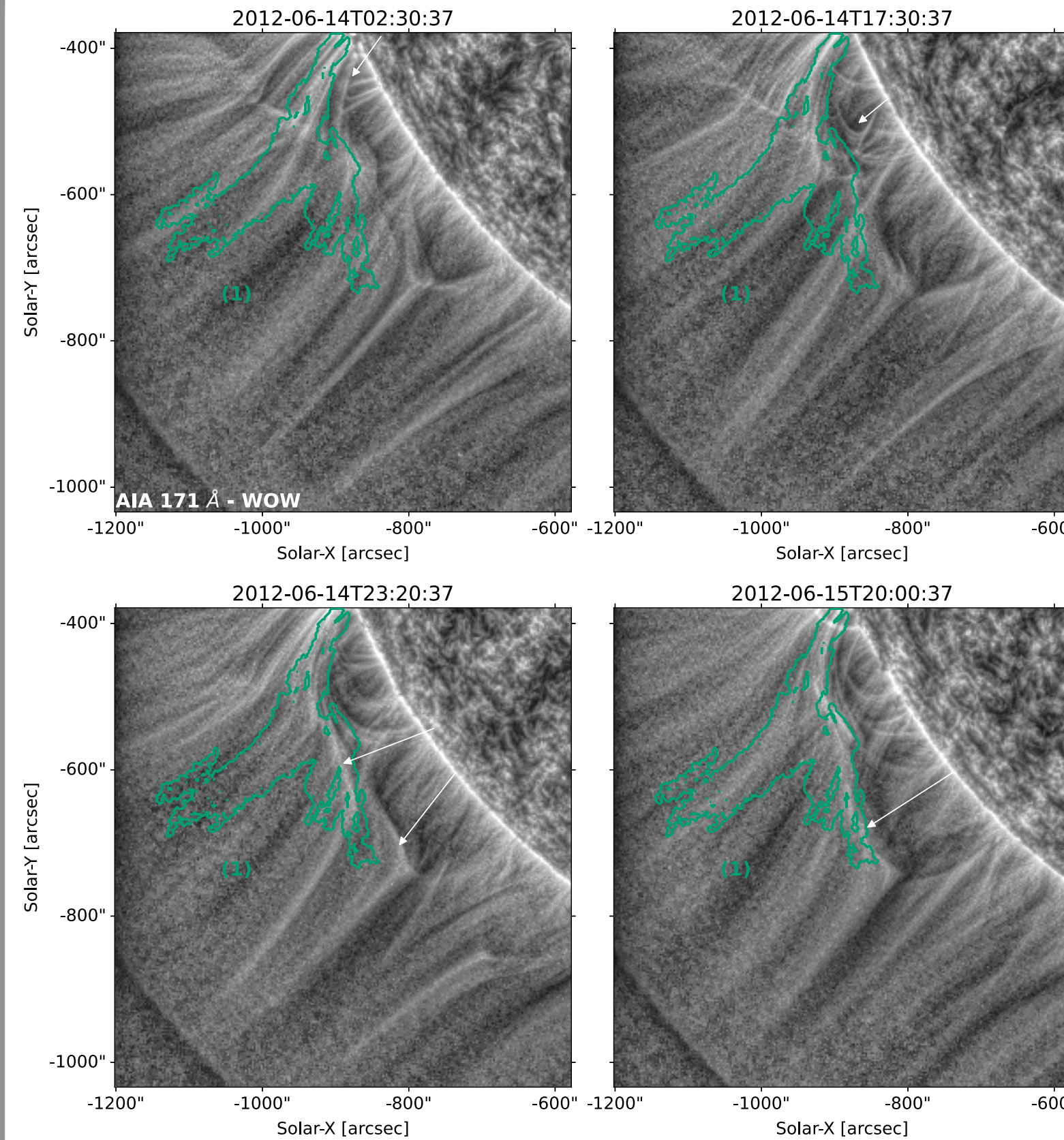
The detection is done in the Fourier space for every pixel of the field-of-view Auchère et al. 2014

- ➔ We do not look for pulsations in a particular structure



- ➔ Detections in two channels (period of 12 hours) + quasi-synchronous recurring behaviour in 304

Pseudo-streamer dynamics: evidence for interchange reconnection

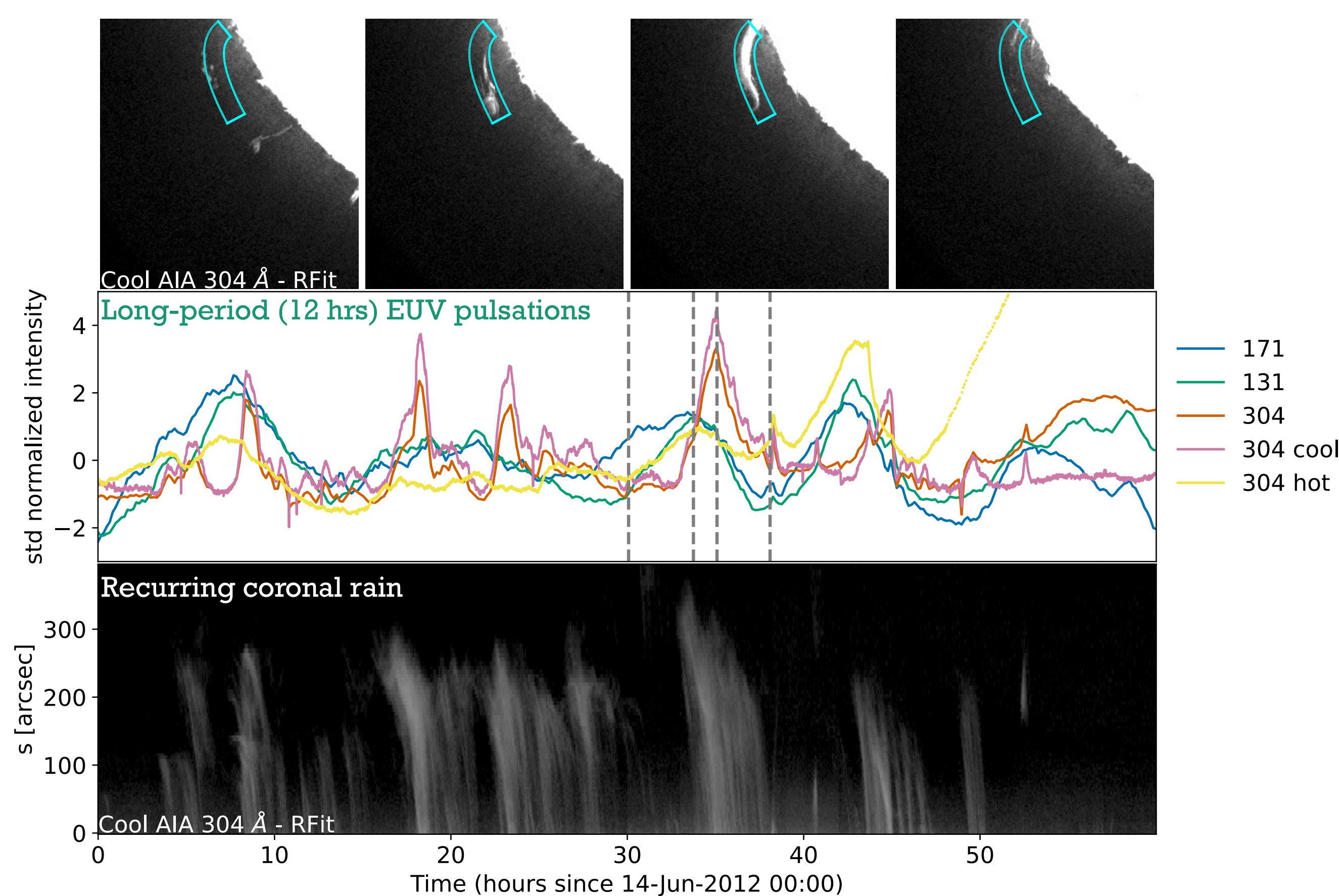


- ➔ Dynamical opening and closing of the closed and open field respectively

- ➔ Interchange reconnection occurs continuously during the entire observation and all along the open-closed boundary

- ➔ Consistent with the observations of Masson et al. 2014 and MHD modeling of Pellegrin-Fracheon et al. 2023

Evidence for thermal non-equilibrium (evaporation & condensation) cycles, first detection in open field!



304 Å components extracted with the RFit method of Antolin et al. 2024

- ➔ Coronal rain showers happen in both the open and closed field

Takeaways and perspectives



Check out the paper! Froment & Masson, Sol Phys 301, 118 (2026)

- ★ Interplay between thermal non-equilibrium and interchange reconnection?

TNE periodically forcing the reconnection?

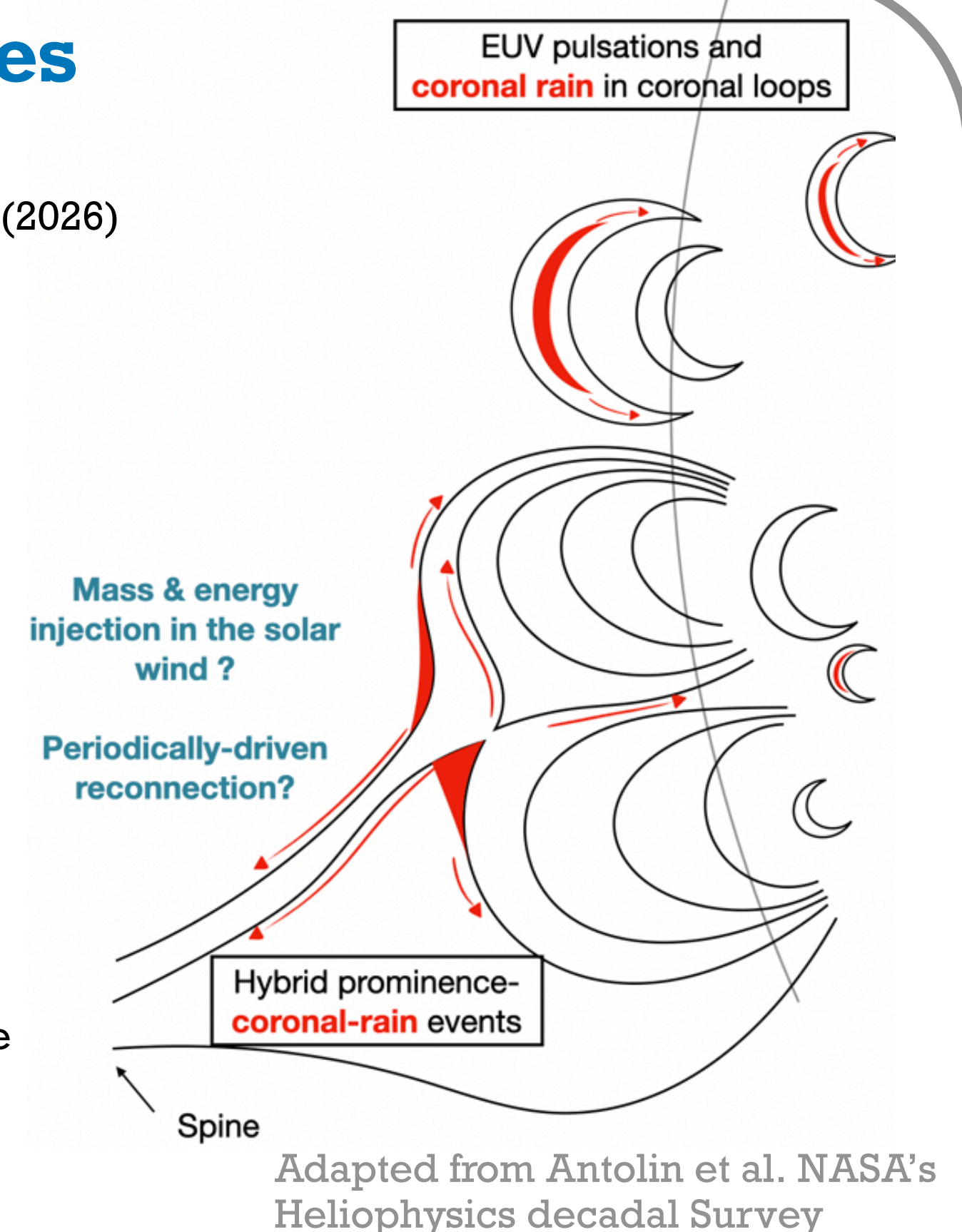
- formation of a filament above an X-point structure followed by coronal rain draining from the magnetic dip Johnston et al. 2025
- TNE cycles driving periodic reconnection in helmet streamers Schlenker et al. 2021

and/or

Reconnection events could destabilise structures that are marginally thermally unstable (close to TNE-prone conditions)

- ★ Injection of mass and energy outward?
- ★ Any solar wind variability driven by this kind of events?

e.g. quasi-periodic density structures injected on a timescale up to a few hours Viall et al. 2010, Viall & Vouridas 2015



Adapted from Antolin et al. NASA's Heliophysics decadal Survey